

Hindi Spoken Conversational System for Healthcare Humanoid

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Abstract—As the world is advancing in robotics, humanising the robot is one of the focused areas. In this work, we showcase our Hindi Spoken Conversational System deployed Mitra robot. The Hindi Spoken Conversational system includes speech recognition, conversational AI and speech synthesis systems.

Index Terms—Automatic Speech Recognition (ASR), Natural Language Processing (NLP), Text-To-Speech Synthesizer (TTS), Humanoid Spoken conversational system.

I. INTRODUCTION

Engineering solutions in the healthcare sector improved the healthcare systems significantly. There are numerous examples of it, from X-ray to Magnetic Resonance Imaging (MRI), laser surgery to stem cell cultivation, etc. Lately, robotic surgery and robots in the healthcare sector have increased significantly. Human-like robots or humanoids are being developed for various domains such as healthcare, military, and retail shops. Human-Robot interaction is the essential want in humanoids. Since speech is humans' most preferred mode of communication, and likewise, vision and navigation are the standard expectation in a humanoid, these features are embedded in it. Recently, there have been numerous companies doing research and development on humanoids and deploying them for various applications.

A Spoken Conversational System is an application that enables human-like conversational interaction with computers, i.e. Humans can communicate with computers in the form of speech or text. Generally, a spoken conversational system has three main components, viz. 1. Automatic Speech Recognition (ASR), 2. Natural Language Processing (NLP), 3. Text to Speech Synthesizer (TTS). The ASR does the job of converting the spoken speech signal into corresponding text. In NLP, the language understanding part understands the input text and identifies the intent of the input sentence. Subsequently, the language generation part replies back with appropriate reply. TTS converts the text into a speech. When these three are in pipeline, it acts as the Spoken Conversational system. Recently there have been more significant improvements in these particular areas because of the boom in machine learning algorithms.

A robot looking like a human having the interaction features like humans do, and at the same time having the ability to

navigate autonomously is a futuristic system that anyone wants to have in his/her home. Building such a system or a robot is a complex task as it involves building many components such as mapping the environment, object detection, navigation, rerouting, face detection and recognition, speech recognition, language understanding, Text-To-Speech systems. Building such components and integrating them to work synchronously in real-time is a challenging task. In this work we attempt to build a speech based conversational system in Hindi using which people can interact with the humanoid in real-time.

The rest paper proceeds as follows. In the second section, the literature review explains the existing similar systems found in the literature. The third section explains a framework for the human-robot interaction. The fourth section explains about the implementations carried out in this regard. Section five concludes the paper and also informs about the possible future works.

II. LITERATURE REVIEW

In literature there are works carried on applying speech to different applications. Even in robotics, the speech is applied for different applications such as education, household and entertainment [1]. In the educational sector, robots like iRobiQ [2], iPal [3] and EngKey [4] helps in childhood education, photo taking, language teaching, gaming and other activities. In healthcare sector, there are robots for different applications such as UVC sterilization [5], Hydrogen Peroxide disinfection [6] and sanitation [7].

From the beginning of research related to the speech recognition in 1950s, the field has come a long way. In the 1980s the HMM based technologies started to popularize because of its performance back then [8]. Later, because of the availability of data and computational resources, the machine learning and deep learning technologies have risen tremendously. ASR systems using RNN, LSTM, and GRU [9] have been implemented. Also, recent transformer based models such as Wav2vec [10] have also been used for speech recognition purpose.

A conversational assistant is a typical example for an AI system. It can mimic human conversation and has a plenty of applications in the domains such as education, e-commerce

and business. Recent advancements have led the tech giants to create world class products such as Apple's Siri, Microsoft's Cortana, Amazon's Alexa, Google's Google Assistant and IBM's Watson. Moreover, open source frameworks and build tools like Rasa provides industry level platforms to build conversational assistants [11]. Due to its support, conversational bots can be built for different languages [12]. Due to the emergence of artificial intelligence, specialized agents were created to assist in particular domains. The open agent architecture (OAA) introduced the influential concept of delegated computing. Then it evolved to multi-agent scenarios [13].

As the name tells Text To Speech Synthesis system, converts the speech into text. Concatenation and statistical parametric synthesis were the state-of-the-art methods for many years in Text-To-Speech synthesis field [14] [15]. WaveNet [16] successfully generated speech-like raw waveform of the audio signal instead of acoustic features. However, at each time-step, the model computes some prediction that made the inferences become slow and computationally expensive. Deep Voice [17], unlike its predecessors, produced an end-to-end deep learning solution which used simple hand engineered features that could be easily generated. Tacotron [18] is also an end-end sequential system. However, as it used Griffin-Lim synthesizer, produced speech sounded robotic. Tacotron-2 [19] is one of the current state-of-the-art model that simplified and improved the results of Tacotron. Mel-spectrogram is converted to audio waveform using Vocoder. WaveNet based architecture produced better results over the traditional parametric vocoders [20], however due to their auto-regressive nature, they suffer from inference speed issues. Generative Adversarial Networks (GAN) based auto-regressive models gave good perceptual quality, however they require a well trained teacher model and methodology to optimize the whole process which is difficult to build for low resource Indic languages. Using Parallel WaveGAN [21] it was able to overcome these issues.

In this quest, a spoken conversation module that combines a Wav2vec-based Hindi Automatic Speech Recognition system, a Hindi Conversational Bot, and Tacotron-Parallel WaveGAN based Hindi Text to Speech Synthesizer is developed. The spoken conversation pipeline module is entirely in Hindi, which is the significant contribution our work.

III. FRAMEWORK FOR HUMAN-ROBOT INTERACTION

In this section, we shall look at a framework for an interactive robot or humanoid.

During the conversation with the robot or humanoid, generally the following steps are involved. They are:

- 1) A person speaks to the humanoid in Hindi language using the web based Graphical User Interface (GUI).
- 2) The audio is recorded through mic and the is transferred to the server.
- 3) Hindi ASR module generates the text output from the audio. This text is then transferred to Rasa server and at the same time displayed on the web browser.

- 4) In the Dialog system, it understands the intent of the sentence, extracts entities, and action is taken correspondingly.
- 5) When a text output comes from the dialog system, the Text-To-Speech synthesis system converts the text to an audio file.
- 6) In the end, the generated audio is played at the terminal/humanoid.

A. Overview of Mitra Robot

Mitra Robot is a product of an Indian startup named Invento Robotics. The robot has autonomous navigation, contextual spoken conversation, and face recognition features. Mitra's spoken conversation module consists of automatic speech recognition system that recognizes the English speech into corresponding English text. The speech recognition system is integrated with the dialog management system. The reply from dialog system is fed into the Text-To-Speech synthesizer. The spoken conversational system is deployed in the proprietary application of Invento Robotics, which includes functions like registration, and face recognition, to name a few.

The robot's navigation system detects the object using a LiDAR sensor. The robot is capable of independent navigation in the inside environment after it has been mapped out. A web interface and mobile application are also available for manual navigation. The wheel's movement is accomplished by two motors linked to a microcontroller. Mitra robot has been already deployed in places like hospitals, clinics, expos, retail shops, and senior care facilities. Invento Robotics is currently focusing on the humanoids in elderly care to assist older people in elderly care or home setup.

B. Spoken Conversation Module

As shown in the Fig. 1, the spoken dialogue module is composed of three primary parts: Automatic Speech Recognition (ASR), Natural Language Processing (NLP), and Text-To-Speech (TTS) synthesiser.

1) *Automatic Speech Recognition:* Automatic Speech Recognition (ASR) is an integral part of the interactive conversation module where we convert speech to text. Conventional approaches like GMM-HMM [22] were initially adopted. However, these models were less accurate and required a huge amount of labelled audio data, which was not practically feasible for all the languages. Later, Deep Learning approaches such as RNN, DNN [23] came into the limelight. Though these models gave better results, they also required large amount of labelled data and computational resources. Recently, self-supervised learning approaches such as Wav2vec 2.0 have yielded good results.

Wav2vec 2.0 [24] is one of the state-of-the-art Automatic Speech Recognition models based on self-supervised training. The architecture has two phases viz., 1. pre-training, 2. fine-tuning. Pre-training is a unsupervised mode where the model learns to extract the contextual representations from unlabeled audio data. in the second phase fine-tuning is carried out where the model is made to map these contextual features to

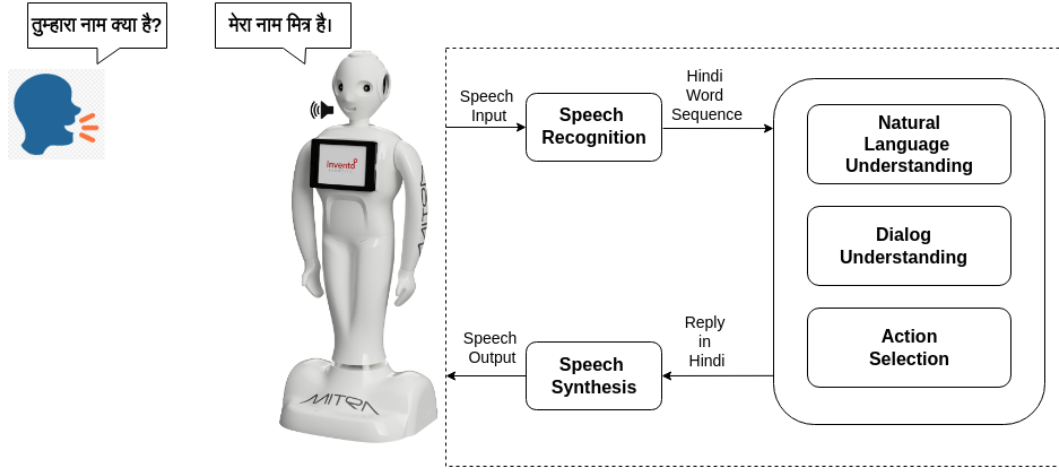


Fig. 1. A framework for Human-Robot Interaction in Hindi, using Mitra robot

corresponding phoneme sequences with the help of labelled audio data. This way of training is considered to be powerful. For low-resource Indic languages such as Hindi, Kannada, Sanskrit, etc., where getting the large amount of labelled audio data itself is practically not feasible [25]. Thus, adopting self-supervised learning is an excellent development in ASR applications for low-resource Indic languages.

2) *Natural Language Processing*: Natural language understanding (NLU) is a subtopic in natural language processing (NLP) that does the understanding part in the dialog task. This enables the computer to understand the human language by converting it into structured data using various algorithms. Ultimately, the intent of the sentence is identified, and entities are extracted which can be appropriately used for the Natural Language Generation, i.e. the reply of the dialog system.

3) *Text To Speech Synthesizer*: A Text-To-Speech (TTS) synthesizer produces a human voice artificially, i.e., to convert a text to a spoken audio [26]. The goal of a good Text-To-Speech system is to not just make machines talk but also to sound them more like humans. Developments in Computer Science and Artificial Intelligence in recent years have made huge impacts in making this possible. Earlier parametric synthesis methods had a significant advantage over concatenation synthesis due to their ability to learn without storing the audio samples in the database, language independence and flexibility in voice characteristics. Hybrid Deep Learning models such as Fast WaveNet [27], Deep Voice 2 [28], Deep Voice 3 [29] and Parallel WaveNet [30] improved the results of their original models. However, they were quite complex and did not produce satisfactory human-like voices.

Tacotron-2, which came as an improvement to Tacotron,

used simpler building blocks. Bi-directional LSTM and convolutional layers were used in the encoder and decoder. Location-sensitive attention encouraged the model to take into account both the focus location and the input features, thus making the model move more consistently through the input avoiding failure modes. To convert the mel-spectrogram generated to an audio waveform, an audio waveform generator (or) Vocoder is used. Parallel WaveGAN do not require a two-stage sequential teacher-student training process. It is easy to train and generate a more natural sound waveform with perceptual quality.

C. Cross-Platform Frameworks for Integration

A spoken conversational system's different components must be integrated into a single module or system. This integration helps encapsulate and run the different modules as a single system without exposing the underlying complexity to the users. There are numerous ways a system can be integrated and run as a single independent system. These ways include:

1. Standalone Desktop Application
2. Web based Application
3. Mobile Application

Different modules such as ASR, Dialog System and TTS can be integrated, and standalone desktop applications can be developed. This integration approach is relatively easy to implement. However, the developed app will run on a single system.

In a web-based application approach, all the modules can be integrated as a web service and follow the traditional client-server architecture. The advantage of building a web application is that it can be hosted locally or globally and accessed over HTTP anywhere around the globe.

The third approach is developing a mobile application similar to desktop application but not limited to any particular mobile platform. Different cross-platform frameworks such as React Native, Flutter, and Ionic can be used to develop cross-platform mobile applications, reducing the development efforts while providing the same accessibility as web-based applications. This cross-platform development allow users to use the same application on their smartphones with different operating systems such as Android and iOS.

IV. INTERACTIVE HINDI SPOKEN CONVERSATION MODULE

In this section, modules built in interactive Hindi spoken conversational system are explained in this section. A schematic representation of the same is given in the figure 1 and explained below.

A. Hindi ASR system

A Wav2vec 2.0 Automatic Speech Recognition model is developed that converts Hindi speech to Hindi text. With the help of pretrained model which is trained on 10,000 Hrs of 23 different Indic languages and then fine-tuned on Hindi IITM data-set for building Hindi ASR. obtained word error rate(WER) of the model is 8.32%.

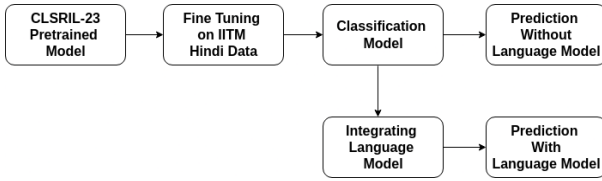


Fig. 2. Dataflow Diagram of Hindi Automatic Speech Recognition

In this architecture context representations are the output from Wav2vec transformer layers. these context representations fed into linear layer followed by a softmax function. The final output contains probability distribution over certain tokens. To convert these probability distributions into text, a decoder is required. Two decoders are used namely Viterbi decoder and KenLM decoder. Viterbi decoder finds the most likely token given the output probability distributions from Wav2vec 2.0 and decodes the next token whereas the KenLM decoder looks at k probable tokens and outputs k probable text sequences, where k is the beam size specified while decoding. KenLM decoder requires a language model to perform decoding.

B. Hindi Dialog system

The Hindi Unicode script is used to write the conversation's inputs and actions. The bot can handle greetings, location-specific information, and suggestions for minor illnesses. In order to create a genuine discussion, the stories are produced utilising potential specified intentions and actions. The DIET classifier in Rasa classifies intents with 99% accuracy which is great for domain specific conversational assistant.

C. Tacotron Hindi TTS

The Text-To-Speech synthesizer built using Tacotron-2 to generate the mel-spectrogram and Parallel WaveGAN to produce the sound waveform from the features for low-level Indic language Hindi. IITM provided 40 hours of TTS dataset is used for training the Tacotron 2 model. The Vocoder; 'Parallel WaveGAN' was also pretrained on the same dataset. During the fine tuning phase, the output from Tacotron2 (mel-spectrogram) was used to generate the output speech.



Fig. 3. Tacotron 2 and Parallel WaveGAN based Hindi Text-To-Speech system

The TTS architecture proposed in this paper was implemented using ESPnet-TTS [31]. End-To-End Text-To-Speech (E2E-TTS) processing toolkit - ESPnet-TTS uses PyTorch as deep engine, supports state-of-the art models such as Tacotron-2 that follows Kaldi style data pre-processing.

D. Web-based Conversational interface

Automatic Speech Recognition, Language understanding and Text-To-Speech modules are integrated together using different web technologies. This integration allows these modules to work as a complete Conversational AI Assistant system. The web technologies and frameworks used for this integration includes HTML, CSS, and Flask framework. The graphical user interface is developed using HTML and CSS as shown in the Figure 4. To record the audio of a speaker, APIs provided by the WebRTC project have been implemented. Using the flask's built in web development server, the Conversational AI System has been hosted on a local server and can be accessed from any computer, tablet or smartphone connected to the same network. The purpose of creating the web based application is to allow users to access the system from across the globe over HTTP.

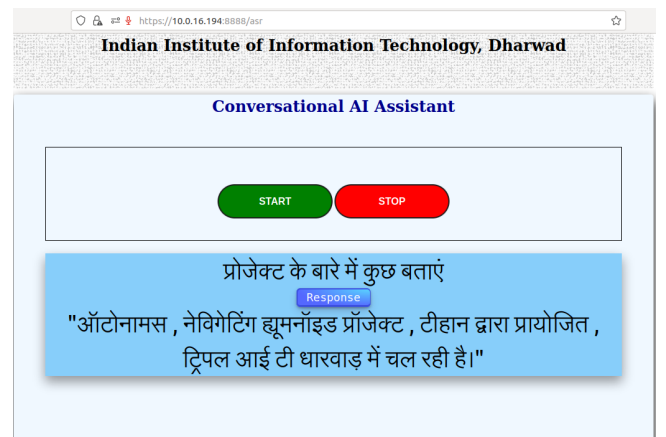


Fig. 4. Hindi Spoken Conversational System User Interface

V. CONCLUSION AND FUTURE WORK

The integrated Spoken Conversational Assistant system makes it a more attractive for the customer. The purpose of robots deployed in the healthcare facilities serves as the people interacts with the robots as talking to a human. The Wav2vec based Hindi ASR pipeline with Rasa chatbot and Tacotron TTS system, yields good real-time performance.

There is a lot of scope for improvements in Conversational system running on robots. The integration of Hindi Conversational system with navigation is a good area to work on. The scope of autonomous navigation integrated with Conversational AI Assistant system has enormous market opportunities. Increasing the domain knowledge of the Hindi Conversational system and making it robust with regard to the noise, recognition and such aspects, greatly increases and grabs the attention of Indian people.

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